Unit 1 - AWS Fundamentals

Cloud Computing | Unit 1

What is Cloud Computing?

Cloud Computing is the delivery of computing services including servers, storage, databases, networking, software, analytics, and intelligence over the internet (the cloud) to offer faster innovation, flexible resources, and economies of scale.

Cloud Service Models

IaaS (Infrastructure as a Service)

Provides virtualized computing infrastructure. Examples: AWS EC2, Google Compute Engine. You manage OS and above.

PaaS (Platform as a Service)

Provides platform for developing and deploying applications. Examples: AWS Elastic Beanstalk, Heroku. You manage only application.

SaaS (Software as a Service)

Delivers software applications over internet. Examples: Gmail, Salesforce, Dropbox. Provider manages everything.

AWS Core Services

- Amazon EC2 - Virtual servers in the cloud
- Amazon S3 - Object storage service
- Amazon RDS - Managed relational database
- Amazon DynamoDB - NoSQL database service
- AWS Lambda - Serverless compute service
- Amazon VPC - Virtual Private Cloud networking
- Amazon CloudFront - Content Delivery Network
- Amazon Cognito - User authentication service

- AWS API Gateway - REST API management
- Amazon SNS/SQS - Messaging services

AWS Global Infrastructure

Regions

Geographic areas with multiple Availability Zones. Examples: ap-south-1 (Mumbai), us-east-1 (Virginia).

Availability Zones

Isolated data centers within a Region. Deploying across multiple AZs ensures high availability.

Edge Locations

Used by CloudFront to cache content closer to users, reducing latency.

AWS Free Tier

AWS Free Tier allows new customers to use certain services for free. It includes 12 months free (like EC2 t2.micro, S3 5GB), Always free services (Lambda 1M requests/month, DynamoDB 25GB), and short-term trials for new services.